

# Invoice AI SaaS Platform

Enterprise Portal (Frontend, Backend, CI/CD, and Private Cloud Infrastructure)

## 1. Platform Architecture & AI Highlights

Designed to combine state-of-the-art Agentic AI with robust, highly scalable infrastructure:



### AI & Intelligence

- **3-Agent Cooperative Network:** Extractor, Evaluator, and Validator agents working in a collaborative LangGraph network.
- **Built-in Anomaly Detectors:** Alerts (math errors, vendor mismatches) generated dynamically during PDF loading.
- **Feedback Training Module:** UI Trainer coordinates overrides directly improve the model's accuracy.



### Platform & Architecture

- **Multiple User Login Personas:** Admin, Auditor, and Loader profiles with strict role-based access.
- **Multi-Source Ingestion:** Pull invoices from cloud/local directories or existing ERP systems.
- **Custom Field Group Tagging:** Pre-upload categorization tags (e.g. #IT, #Contractors) to segment data.
- **Audit Records & Status Lifecycle:** Traceable Paid/Rejected transitions logging auditor user details.
- **Semantic Chat Explorer:** Natural language database query tool that cites source PDF pages.



### Cloud & Infrastructure

- **Horizontal 1-100+ User Scaling:** Stateless API layers and database tenant pools scaling smoothly.

- **Environment Cost Optimization:**  
Omit optional Dev sandboxes to save recurring cloud subscription costs.
- **Dynamic Dashboard Analytics:**  
Custom stats, trend charts, and tag-based custom field summaries.

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Slide 1/7 | Platform Overview

## 2. Screen 1: Dashboard Panel (Tab 1)

High-level financial summaries, metrics, and active Salesforce/ERP integration status logs at a glance.



### ✓ MVP Scope (POC Match)

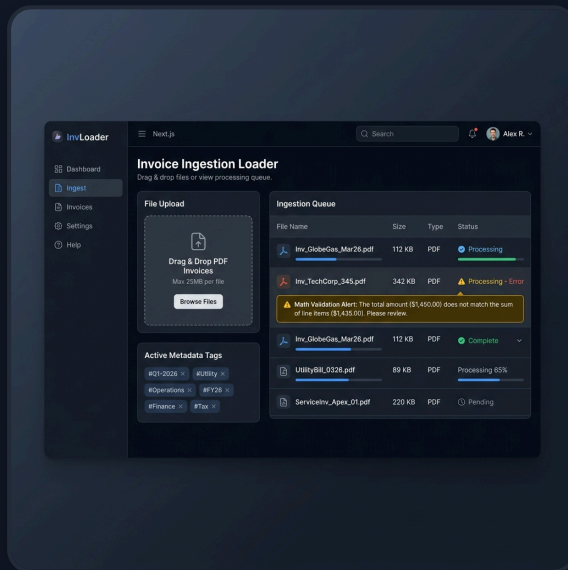
- Total spend tracking metrics cards and status counters.
- Global filter controls bar (date ranges and vendor multi-select dropdowns).
- Historic invoice list metrics and trend charts.

### ⚠ Future Placeholders

- **Live ERP Integration:** Active Salesforce, SAP, and QuickBooks sync status connectors widgets.
- Direct ledger summaries export triggers.

### 3. Screen 2: Ingestion Loader (Tab 2)

Core upload interface featuring row-level tag classification and non-blocking anomaly queues.



#### ✓ MVP Scope (POC Match)

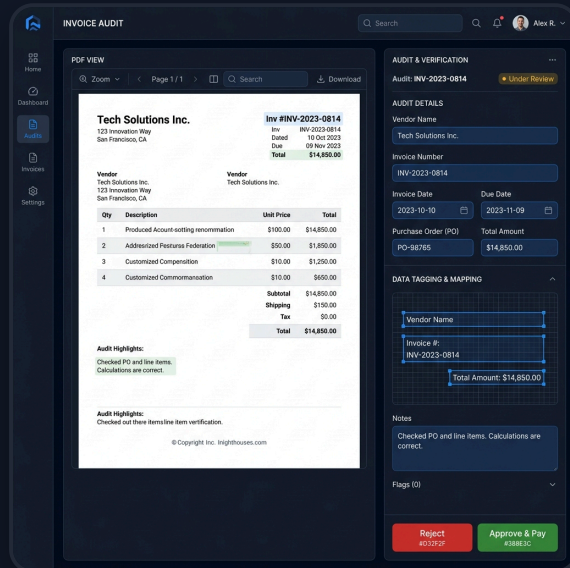
- Drag-and-drop secure file uploader to Azure storage.
- **Row-level checkboxes** to apply tags (e.g. #IT, #Contractors) **before uploading**.
- No-block ingestion: Uploads process fully, generating warning alerts instead of rejections.

#### ⚠ Future Placeholders

- Ingestion history query archives sub-tab.

## 4. Screen 3: Auditor Console (Tab 3)

Split-screen layout displaying OCR visual coordinate extractions, values overrides, and alert dismissal hooks.



### ✓ MVP Scope (POC Match)

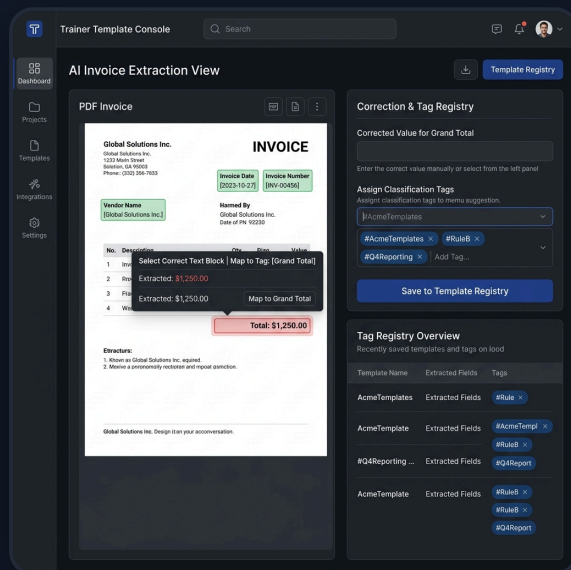
- Split-screen layout: PDF layout on the left, editable metadata values on the right.
- Status lifecycle: Mark invoices as Paid (stamps PAID) or Rejected (stamps REJECTED).
- Audit logging: Captures and shares timestamps and paid/rejected auditor user details.

### ⚠ Future Placeholders

- **Interactive Alerts Desk:** View alerts, match anomalies to coordinates, and click to remove/dismiss individual alerts.

## 5. Screen 4: Trainer Console (Tab 4)

Coordinate adjustment interface to create spatial templates and override AI extraction rules.



### ✓ MVP Scope (POC Match)

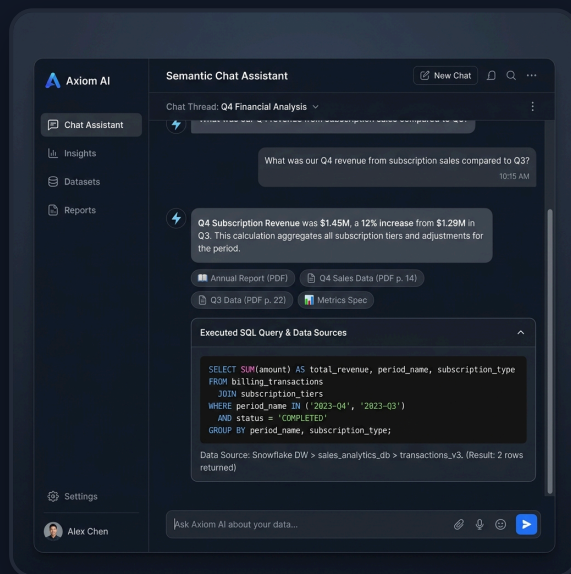
- OCR coordinates overlay grid highlighting AI text extraction boundaries.
- Click-to-Correct bounding boxes to manually adjust target positions.

### ⚠ Future Placeholders

- **Template Registry:** Bind corrected offsets to category rules to bypass AI model steps on future matching layouts.

## 6. Screen 5: Semantic Query Chat (Tab 5)

Conversational chat console querying database aggregates and outputting cited page references.



### ✓ MVP Scope (POC Match)

- Conversational bubble chat log interface.
- Clickable citation chips linking back to source PDF pages.

### ⚠ Future Placeholders

- **SQL Explainer Dropdown:**  
Collapsible SQLModel query logs to check LLM database execution paths.

# 7. Project Timelines & Environmental Costs

## Development Implementation Roadmap

| Roadmap Component                    | Timeline / Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Billing Model                     |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Phase 1: Core MVP (POC Match)        | 4 Months — Delivers core MVP screens: <ul style="list-style-type: none"><li>▪ <b>Dashboard:</b> Spend bento cards, date/vendor filters, and trends.</li><li>▪ <b>Ingestion Loader:</b> Pre-upload tagging (#IT/#Contractors) &amp; non-blocking alerts.</li><li>▪ <b>Auditor Console:</b> Split review &amp; metadata form, Paid/Rejected stamp logs.</li><li>▪ <b>Trainer Console:</b> OCR bounding box overlays &amp; correction canvas.</li><li>▪ <b>Semantic Chat:</b> Conversational queries citing source PDF pages.</li></ul> | ₹2,00,000 (2 Lakh) / Month        |
| CI/CD & Cloud Setup (Optional)       | One-time configuration of deployment pipelines, secure environments, and cloud infrastructure setup.                                                                                                                                                                                                                                                                                                                                                                                                                                 | ₹2,00,000 (2 Lakh) One-time Setup |
| Phase 2: Full Integration (Optional) | 2 to 3 Months — Delivers Salesforce/SAP connectors, settings panels, and template learning registry.                                                                                                                                                                                                                                                                                                                                                                                                                                 | ₹2,00,000 (2 Lakh) / Month        |

## Azure Recurring Cloud Costs (Per Environment)

Scales dynamically at approx. ₹5 INR per invoice processed.

| Target Environment | Deployment & Scaling Details                                                                                                   | Estimated Monthly Cost (INR) |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Production (Prod)  | High Availability database, Redis queues, and ACA horizontal container nodes.                                                  | ₹30,000 – ₹55,000 / Month    |
| UAT & Dev          | Staging sandbox for QA and testing.<br><i>*Note: Dev environment is optional and can be omitted to save operational costs.</i> | ₹15,000 – ₹20,000 / Month    |



Slide 7/7 | Project Deliverables